recently been developed. It is hoped that the study of hardness measurement may assist in developing a standard method of testing. The object of this investigation is to determine to what extent agreement exists between the results obtained by methods which are apparently in no way related to each other. In this preliminary work no attempt has been made to do more than to obtain comparative results for a series of metals which would be sufficient to show the performance of the individual instruments. Consequently no determinations have been made of the elastic constants or other properties of the metal, such as action under alternating stresses, impacts, cold bending, torsion, tension, or compression, although it is evident from the nature of the problem that these constants and properties are necessary for the interpretation of results.

As early as 1722 Reamur originated a test for hardness which depends upon the principle that a harder metal scratches a softer one. Since that time various forms of sclerometers have been introduced, all of which depend upon this principle. To Brinell we owe the introduction of a method of hardness measurement which can be easily applied and the results of which can be expressed in definite physical units.

Brinell originated this method in 1900 while chief engineer and manager of the Fagersta Iron & Steel Works in Sweden. The method as he developed it is based upon determining the resistance offered to indentation by a hardened steel sphere when this sphere is placed on the material under investigation and subjected to a given pressure. He defined the hardness numeral as the ratio of the pressure on the sphere to the area of the spherical indentation produced.[1]

[1] H. Hertz first defined hardness as the mean pressure which exists at the center of the indentation when the material under test has just reached its elastic limit. This definition he applied only to bodies which have surfaces of such a form that a circular indentation results when they are pressed together. Hertz confined his hardness measurements to glass and similar brittle substances in which the attainment of the elastic limit is denoted by the appearance of a circular fissure around the contact area. Such a fissure does not necessarily denote that the material at the center of the indentation has passed its elastic limit. Hertz has shown that it is impossible to apply his definition to metals, because there are no means of telling when the elastic limit is reached at the center of the indentation. (For the mathematical development of the Hertz equations, see H. Hertz Gesammelte Werke, vol. 1, p. 155 ff., or Annalen der Physik; vol. 14, 1904, p. 153 ff. For discussion of the limitations of measuring hardness according to the Hertz definition, see vol. 51, Zs. des Ver. Deutscher Ingenieur Prüfverfahren für Geharteten Stahl under Berücksichtigung der Kugelform Prüfungs ergebnisse Elastische und Bleibende Form anderungen. R. Stribech; vol. 52, Zs. des Ver. Deutsch. Ingenieur, Unter suchungen über Harte Prüfung und Harte. E. Meyer. C. Bach. Elasticitat und Festigkeit, verlag von Julius Springer.)